No way to linguistic compositionality without Duality of Patterning

Joana Rosselló[*1]

[*]Corresponding Author: joana.rossello@ub.edu

[1]Dpt. of Catalan Philology and General Linguistics, University of Barcelona, Spain

Claims that compositionality exists in non-human animal communication systems (ACS) rely on weakened or modified notions of the concept and do not provide a plausible evolutionary precursor of linguistic compositionality. This paper argues that such claims arise from a misinterpretation of Duality of Patterning (DoP) and from conflating compositionality with more general forms of complex semanticity (see Townsend et al., 2018; Zuberbühler 2019, 2020; Schlenker et al., 2023).

1. DoP is a unitary property

Within Hockett's Design Features of Language (DFL), DoP is a unitary property: minimal meaningful units are composed of meaningless but contrastive elements. However, recent approaches in language evolution, based on an extended construal of DoP, split DoP into combinatoriality and compositionality (Amici et al., 2024; de Boer, Sandler & Kirby, 2012); Gontier et al., 2024; Pleyer et al., 2022). This partition obscures the internal logic of the design features and misrepresents Hockett's view. Combinatoriality, understood as the combination of meaningless elements into minimal meaningful units, is already captured by DoP in its core sense (Hockett 1958a: 577; see Ladd (2012)). Compositionality, by contrast, corresponds to DoP in its core sense + Productivity (Hockett 1958a; 1958b: 150–151). Thus, compositionality is implicit in Hockett's original proposal. Treating it instead as an independent feature —crucially, in the absence of pleremes—, would make it impossible to express that linguistic compositionality can arise only in a semantic system that is both discrete and productive.

2. Complex semanticity rather than compositionality

Empirical claims of compositionality in ACS instead reveal forms of complex semanticity. These claims typically rely on relaxed interpretations of the Fregean definition, particularly regarding syntax. In studies of bird calls (Suzuki, 2016, 2019, 2022), sequential order is taken to suffice for syntax, despite the absence of hierarchical structure (Bolhuis et al. 2018a, 2018b). In great ape vocalizations (see Berthet et al. 2025 on bonobos; Girard-Buttoz 2025 on chimpanzees), increasingly complex semantic combinations are posited, but these differ fundamentally from the compositional ones obtained via the simpler binary merge characteristic of human language. Moreover, such accounts presuppose meaningful subparts, yet "meaning" in these systems is not reducible to functional referentiality (Rosselló & Zhang, 2023) but is instead sheerly externalist, tied to contexts or situations rather than to discrete symbolic units (i.e., morphemes/words). The resulting interpretations are holistic (e.g., "let's stay together", "I'm excited") and lack reference to third entities.

3. An alternative evolutionary view

From an evolutionary perspective narrowly confined to descent with modification, diverse authors —including Hockett himself— contend that holistic calls likely preceded DoP and therefore, discreteness internal to the signals (see for instance Wray (2000) and Sandler (2011); and the critical perspectives of Bickerton (2007) and Lillo-Martin (2020), respectively). However, the transition from externally grounded signals to internally structured, compositional systems remains unexplained. Animal vocalizations are graded, biologically constrained, and non-conventional, whereas words are discrete and conventional, i.e., symbolic. This discontinuity suggests that linguistic compositionality cannot emerge directly from ACS.

An alternative view is proposed in which the invention of words is the crucial step. Words can be understood as a cultural technology that grafts meaning onto the sound sequences yield by vocal production learning —phonological syntax, Marler (1998)—, an evolutionary analogy (Jarvis, 2019). Because phonological sequences are discrete and meaningless, they can support conventional, abstract meanings —the "honest fakes" (Knight, 2008, 2018). In this sense, *Homo sapiens* is a bio-cultural hybrid in a literal sense since words are prostheses in human brains: DoP constitutes the interface between biological capacities and cultural conventions. This framework, reminiscent of Studdert-Kennedy (2000), explains why linguistic compositionality depends on DoP and why it cannot be derived from systems that lack it.

The proposal has broader implications. It predicts that compositionality in other domains (Nefdts & Potts, 2024; Gontier et al., 2024; Vicente et al., 2023; Dautriche et al. 2022) cannot be transferred to language, aligns with neurobiological evidence linking phonological and syntactic processing (Hickok, 2017; Matchin & Hickok, 2020), and supports scenarios in which words combined with gestures enabled the emergence of declarative communication. Overall, compositionality is not a gradual extension of animal communication, but a consequence of the unique integration of discreteness, conventionality and productivity made possible in communication by Duality of Patterning.

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